# HAO ZHOU

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## ACADEMIC APPOINTMENT

**Assistant Professor** 2026.08 -
Department of Computer Science, University of Texas at Dallas

## RESEARCH AREAS

AI-Powered Mobile and Wearable Systems · Human-Centered AI · AI for Health

## EDUCATION

**The Pennsylvania State University**
Doctor of Philosophy in Computer Science and Engineering 2021.08 - 2026.05

**The University of Mississippi**
Master of Science in Computer Science 2019.08 - 2021.05
Bachelor of Science in Computer Science 2016.08 - 2019.05

## HONORS & AWARDS

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| **2025** | Penn State Alumni Association Dissertation Award |
| **2025** | Samsung Research America President Award |
| **2025** | ACM MobiSys Rising Stars |
| **2025** | UIST Special Recognition for Belonging and Inclusion Award |
| **2024** | ACM MobiCom Student Travel Grant |
| **2023** | Entrepreneurial Lead in NSF I-Corps Program |
| **2023** | ACM/IEEE IoTDI Best Paper Award for Edge IoT AI |
| **2023** | Microsoft Research Asia Stars of Tomorrow |
| **2023** | Penn State International Student Travel Grant |
| **2022** | Penn State Outstanding Teaching Assistant Award |

## PUBLICATIONS

**ICML 26** *Physiology-Aware Masked Cross-Modal Reconstruction for Biosignal Representation Learning.*
**Hao Zhou**, Simon A. Lee, Cyrus Tanade, Keum San Chun, Juhyeon Lee, Migyeong Gwak, Megha Thukral, Justin Sung, Eugene Hwang, Mehrab Bin Morshed, Li Zhu, Viswam Nathan, Md Mahbubur Rahman, Subramaniam Venkatraman, Sharanya Arcot Desai

**MobiCom 26**
*Exploring the Feasibility of Full-Body Muscle Activation Sensing with Insole Pressure Sensors.*
**Hao Zhou**, Mahanth Gowda

**ICLR 26**
*HiMAE: Hierarchical Masked Autoencoders Discover Resolution Specific Structure in Wearable Time Series.*
Simon Lee, Cyrus Tanade, **Hao Zhou**, Juhyeon Lee, Megha Thukral, Minji Han, Rachel Choi, Md Sazzad Hissain Khan, Baiying Lu, Migyeong Gwak, Mehrab Bin Morshed, Viswam Nathan, Md Mahbubur Rahman, Li Zhu, Subramaniam Venkatraman, Sharanya Arcot Desai

**Nature Comm.**
under review
*ECG-informed pretraining enables precise cardiac assessment from wearable photoplethysmography.*
**Hao Zhou**, et al.

**Nature Comm.**
under review
*A compact on-device foundation model enables real-time physiological screening on wearable devices.*
Simon A. Lee, **Hao Zhou**, et al.

**NeurIPS 26**
TS4H
*Towards On-device Foundation Models for Wearable Signals.*
Simon Lee, Cyrus Tanade, **Hao Zhou**, Juhyeon Lee, Megha Thukral, Minji Han, Baiying Lu, Sharanya Desai

**ICASSP 26**
*A Personalized Real-time Proactive Voice Memory Assistant.*
**Hao Zhou**, Md Mahbubur Rahman, Simon A Lee, Baiying Lu, Juhyeon Lee, Cyrus Tanade, Megha Thukral, Md Sazzad Hissain Khan, Samsad Ul Islam, Subramaniam Venkatraman, Sharanya Arcot Desai

**MobiSys 25**
Rising Star
*Rethinking Inexpensive Wearables in the Era of AI: From Motion Analytics to Mobile Health.*
**Hao Zhou**

**ICASSP 25**
*Know Your Heart Better: Multimodal Cardiac Output Monitoring Using Earbuds.*
**Hao Zhou**, Md Mahbubur Rahman, Mehrab Bin Morshed, Yunzhi Li, Md Saiful Islam, Larry Zhang, Jungmok Bae, Christina Rosa, Wendy Berry Mendes, Jilong Kuang

**UbiComp 25**
*WindDancer: Understanding Acoustic Sensing under Ambient Airflow.*
Kuang Yuan, Dong Li, **Hao Zhou**, Zhehao Li, Lili Qiu, Swarun Kumar, Jie Xiong

**UIST 25**
*SignGlass: First-Person View Comprehensive and Generalizable ASL Translation Using Wearable Glasses.*
Yongxiang Cai, Taiting Lu, Zhenghao Li, **Hao Zhou**, Kenneth DeHaan, Xuhai Xu, Mahanth Gowda, Yincheng Jin

**CHI 25** *BallistoBud: Heart Rate Variability Monitoring Using Earbud Accelerometry for Stress Assessment.*
Md Saiful Islam, Md Mahbubur Rahman, Mehrab Bin Morshed, David J. Lin, Yunzhi Li, **Hao Zhou**, Wendy Berry Mendes, and Jilong Kuang

**MobiCom 24** *Rethinking Orientation Estimation with Smartphone-Equipped Ultra-Wideband Chips.*
**Hao Zhou**, Kuang Yuan, Mahanth Gowda, Lili Qiu, Jie Xiong

**IoTDI 24** *ASLRing: American Sign Language Recognition with Meta-Learning on Wearables.*
**Hao Zhou**, Taiting Lu, Kenneth DeHaan, Mahanth Gowda

**UbiComp 24** *SmartDampener: An Open Source Platform for Sport Analytics in Tennis.*
Runze Liu, Taiting Lu, Shengming Yuan, **Hao Zhou**, Mahanth Gowda

**MobiCom 23** *SignQuery: A Natural User Interface and Search Engine for Sign Languages with Wearable Sensors.*
**Hao Zhou**, Taiting Lu, Kristina Mckinnie, Joseph Palagano, Kenneth DeHaan, Mahanth Gowda

**IoTDI 23** *One Ring to Rule Them All: An Open Source SmartRing Platform for Finger Motion Analytics and Healthcare Applications.*
**Hao Zhou**, Taiting Lu, Yilin Liu, Shijia Zhang, Runze Liu, Mahanth Gowda

**TIOT 23** *I Am an Earphone and I Can Hear My User's Face: Facial Landmark Tracking Using Smart Earphones.*
Shijia Zhang, Taiting Lu, **Hao Zhou**, Yilin Liu, Runze Liu, Mahanth Gowda

**NeurIPS 23** Workshop *Backdoor Threats from Compromised Foundation Models to Federated Learning.*
Xi Li, Songhe Wang, Chen Wu, **Hao Zhou**, Jiaqi Wang

**UbiComp 22** *Learning on the Rings: Self-Supervised 3D Finger Motion Tracking Using Wearable Sensors.*
**Hao Zhou**, Taiting Lu, Yilin Liu, Shijia Zhang, Mahanth Gowda

## INDUSTRY EXPERIENCE

**2025.07 - 2026.07** Digital Health Team, Samsung Research America

**2024.05 - 2024.08** Digital Health Team, Samsung Research America

**2023.05 - 2023.08** Microsoft Research Asia - Shanghai

## PATENTS

**2026** Pending *Cross-Modality Encoding and Reconstruction of Related Physiological Signals.*
**Hao Zhou**, Md Mahbubur Rahman, Cyrus Tanade, Subbu Venkatraman, Sharanya Desai

**2026** *A Personalized Real-time Proactive Voice Memory Assistant.*
Pending **Hao Zhou**, Md Mahbubur Rahman, Subbu Venkatraman, Sharanya Desai

**2026** *Efficient On-Device Learning Frameworks for Wearable Sensor Signals.*
Pending Simon A. Lee, **Hao Zhou**, Minji Han, Rachel Choi, Juhyeon Lee, Sharanya Desai

**2025** *Cardiovascular Biomarker Estimation Using PPG And BCG from a Wearable Device.*
Mehrab Bin Morshed, Md Mahbubur Rahman, Li Zhu, **Hao Zhou**, Larry Zhang

**2024** *Platform for finger motion analytics and healthcare applications.*
Taiting Lu, **Hao Zhou**, Mahanth Gowda, Yilin Liu, Shijia Zhang, Runze Liu

## MEDIA COVERAGE

**2026** xMAE: Samsung News / Machine Brief / RichlyAI

**2025** Penn State Alumni Association Dissertation Award: Penn State News

**2023** OmniRing: Hackster / DeepTech

## INVITED TALKS

**2026.10** Auburn CS

**2026.03** UW-Madison CS

**2026.02** UT Dallas CS

**2026.02** UMBC CSEE

**2026.02** UAB CS

**2025.11** Penn State EE

**2023.05** Microsoft Research Asia

## TEACHING

**Instructor @ UT Dallas**

- CS 4390: Computer Networks SP 2027
- CS 4485: Computer Science Project FA 2026

**Teaching Assistant @ Penn State**

- CMPEN/EE 362: Communication Networks SP 2025
- CSE/EE 559: Wireless and Mobile Sensing in the age of IoT FA 2024
- CMPEN 462: Wireless Communication Systems and Security SP 2022, SP 2024

## SERVICE

**Program Committee**

- ACM MobiCom S3 Workshop 2024
- ACM MobiCom Artifact Evaluation 2024
- ACM MobiSys Artifact Evaluation 2024

**Invited Reviewer**

- ACM UbiComp/IMWUT 2022 - 2027
- IEEE Transactions of Mobile Computing 2024 - 2027
- npj Nature Digital Medicine 2026 - 2027
- ACM CHI 2026 - 2027
- ICLR 2024 - 2027
- ICML 2026 - 2027
- NeurIPS 2026 - 2027
- TMLR 2026 - 2027
- AAAI 2027
- ACM Transactions on Sensor Networks 2024
- Journal of Intelligent Manufacturing 2022